⚠ **V13 / V14 SCOPE UPDATE — REVISION TO v4.6 IN PROGRESS.** Whitepaper **v4.6**: the current-state summary above and the correction banners are current (October 2026); the sections below remain the **v4.5** base architecture (March 2026, consensus, monetary policy, four pillars), with each delta noted inline. It predates the 2026-05-18 V13 scope decisions and predates the V13 RC1 release. **For the canonical current scope**, read docs/V13_PUBLIC_SCOPE_UPDATE.md, docs/V13_GOLD_VAULT_GOVERNANCE_GATES.md, docs/V13_POPC_ESCROW_AUTO_ACTIVATION_GAPS.md and docs/V13_DTD_FLIP_12100_AUTOMATIC.md. The four critical updates relative to v4.5 are:

1. **V13 confirmed at block 12,000** — full cASERT equalizer range **E7-H35** (not the H20 ceiling described in §PILLAR 1 below), DTD reward cooldown 5 → 6, future-timestamp drift cap 60 s → 30 s (**NTP synchronisation strongly recommended post-V13**), Beacon Phase II-A. RC1 metadata is signed (release_status = signed_metadata_only); binaries upload is the next manual operator step.
2. **V13 verified automatic flip at block 12,100** — the V11 Phase 2 lottery cadence transitions from 2-of-3 bootstrap to 1-of-3 permanent without any operator action, restart, RPC, Beacon notice or config flag. Six audit gates GREEN on main.
3. **V13 anti-dominance DTD gate at block 12,100** — bundled with the cadence flip, any miner whose share of the previous **288 blocks is ≥ 10 %** is automatically excluded from the DTD reward eligibility set until their rolling share drops below 10 %. A companion **SbPoW-activity gate** admits only miners that have produced at least one SbPoW-signed block (most recent block at height ≥ 7,100), excluding dormant pre-SbPoW addresses. Dominant miners keep their 50

% miner share on every block they produce; they only miss the redistributed 25 % + 25 % network-side share on DTD-triggered blocks. The exclusion is reversible per-block, no admin authority, no slash, no permanent ban. Independent of the recent-winner cooldown (which continues to apply). See docs/V13_DTD_DOMINANCE_GATE.md.

4. **V14 at block 15,000 — Phase I governance + PoPC automation block** — activated at block 15,000 (now live on mainnet). **Activates:** Gold Vault Phase I governance (90 % miner approval over a 67-block window + 10 % quality boost or single-signer veto from developer / genesis, silence = accept, single destination = genesis miner, cap 1,000 SOST per spend / 5,000 SOST per rolling 1,008-block week); the single-model PoPC bond migrated out of popc_registry.json into deterministic chain-state (**note:** the PoPC redesign moved the base single-bond model activation to **V15 / block 25,000**, **— superseded: the DTD does NOT require a PoPC bond at any height. The mainnet gate ships deactivated (DTD_POPC_GATE_CONSENSUS_ACTIVE = false) and the lottery is permissionless**; the native SOST bond is the only collateral and the only slashable asset, slashed only on a consensus-defined bond failure); Beacon II-A real operator pubkey replaces the placeholder (canonical fingerprint bbb560e3ec86114a59762d467d645c88cfe0497a8f7ca542c973e2e0def8186b = sha256 of the lowercase uncompressed pubkey hex; recompute against src/beacon.cpp to detect a substituted key); threshold sync 95 % → 90 %. **Conditional in V14:** atomic swap HTLC activates only if the libwally-core sprint + end-to-end testnet pass before block 15,000 (this ultimately activated at V14.5 / #16,000; relay/policy flag-day #17,000); ⚠ **historical:** the former PoPC Model B (gold escrow via an Ethereum SOSTEscrow contract, including any "supervised mode") is **superseded** by the single native SOST bond — there is no gold-escrow PoPC in the current design; gold is an optional reward boost only, never collateral, never escrow, never slashed.
5. **Threshold revised: 75 % → 95 % → 90 %** — the Gold Vault miner-signaling supermajority finalised at **≥90 %** over a 67-block window (~12 h, ceil = 61 blocks). The live normative threshold in §Protocol Summary and §Decentralization now reads ≥90 %; any "≥95 %" that remains there is shown only as historical context (the V6-era V14-deferred design). PDF v4.6 regeneration is an operator step.

Memory-Lock per-instance, mentioned in earlier drafts as a second anti-pool defense, was **rejected outright** after numerical analysis (penalises small miners proportionally more than large rigs). SbPoW (active since block 7,100) remains SOST's only anti-pool defense.

WHAT THE PROTOCOL DOES TODAY — read this before the sections below.

This whitepaper still describes the **original** gold/PoPC economics in several places. They were superseded on chain:

- **Block reward split: 50% miner / 50% DTD** since **V15, block #25,000.** The Gold Funding Vault and the PoPC Pool receive **no new emission** —

their historical balances stay on chain and auditable, but nothing new flows in.

- **The DTD is permissionless.** No PoPC bond, no whitelist, no registration: mining one valid block is what makes an address eligible. From **#30,000** (V16) eligibility additionally requires having mined within the last **288** blocks.
- **#30,000** activates V16: DTD-normal eligibility changes and the **DTD Accumulated Reward V2** becomes an independent, node-gated, PoW-weighted draw (first V2 draw **#30,186**). The same activation brings on-chain the **native-asset layer** (asset genesis / issue / transfer / burn, conservation + supply-cap safety, reorg-safe index), **four tokenization modalities** (Tokenize, Auction, Draw, Project Funding) and a **SOST-layer DEX** — all **admin-gated at consensus (S14 RESTRICTED DEVELOPER MODE) with public access disabled** (technical availability is not regulatory authorization; ordinary SOST transfers are unaffected). It is **one** activation, automatic, by block height. Current release: **V30000** (tag v30000) — all three binaries (node, miner, cli) changed.
- **SOST is not backed by gold.** The metals reserve is a Foundation-held reserve with published policy; it is not a peg, not a redemption right and not a claim on the protocol. Reserve status.

HOW THE DTD WORKS · DETERMINISTIC & AUDITABLE

In plain terms, **deterministic token distribution** means the split is computed from **fixed, reproducible rules** — never chosen by anyone.

> If two people look at the **same blockchain data**, they must get **exactly the same result.**

It does **not** depend on who runs the calculation, which phone or computer, the browser, the internet speed, or any manual decision.

✓ Deterministic

Rule: “Distribute 100 SOST among every eligible address of the last 288 blocks by this exact formula.” If there are **193** eligible and the formula is fixed, **every** node that computes it gets the same **193** eligible and the same split.

✗ Not deterministic

Rule: “If an RPC call fails, skip that block.” A slow phone could skip more blocks than a laptop and read **111** eligible while another reads **193** — exactly the kind of defect SOST fixed.

same blocks + same rules + same state = same result

Auditable by anyone. Because the outcome is pure consensus data, anyone can re-derive the eligible set and the winner straight from the public chain and confirm it matches — via the explorer’s getlotteryaudit (per jackpot) or by recomputing the eligibility set over the last 288 blocks. You never have to trust a website or a person.

ADAPTIVE NODE PARTICIPATION FOR NORMAL DTD — FUTURE DESIGN

DESIGN PROPOSAL · NOT ACTIVE · NO THRESHOLD · NO ACTIVATION HEIGHT · NO CONSENSUS CHANGE

Rationale. Independent validating nodes are what make the chain hard to censor and hard to mislead. Jackpot V2 already makes a **bound node identity** (NODE_BIND + heartbeats) a condition for the jackpot. Extending that condition to the normal DTD would convert more miners into node operators, but applying it while the network is small would raise the cost of entry for new miners. The proposal is therefore **adaptive**: the requirement could appear only once the network is demonstrably mature.

Current state (V30000, unchanged by this note)

MINER REWARD	valid mining; never depends on NODE_BIND
DTD NORMAL	every block; mined within the last 288 blocks; recent-producer cooldown (producers of the previous 6 blocks skipped), dropped by a fallback if it would leave the set empty; anti-dor last 288 blocks ≥ 10% excluded) only when ≥ 11 distinct miners are active; uniform pick; NO NODE_BIND; NO heartbeats
DTD JACKPOT V2	every 288 blocks, independent draw (separate domain-tagged seed); ≥ 3 SbPoW blocks in the last 2,016; NODE_BIND; heartbeats (ramping to 3 of the last 4 epochs); pick weighted by PoW blocks

Proposed transition

EARLY NETWORK	normal DTD open to recent miners; NODE_BIND optional for it
MATURE NETWORK	normal DTD MAY additionally require an active bound node identity, only after objective maturity conditions are satisfied

```
ALWAYS          the gate affects DTD eligibility only -- never block validity,
                never the miner's base block reward
```

Measuring maturity

Not by NODE_BIND count alone. Candidate inputs: distinct active miners; distinct active bound node identities; the share of active miners with an active bound node; persistence over time; network stability; observable node diversity; a minimum duration before activation; protection against temporary spikes. **Thresholds are TBD** and must come from simulation and incentive analysis — no number is proposed here.

Manipulation risk and required properties

If activation depended only on a simple count of miners with a bound node, a miner could rationally decline NODE_BIND to keep the stricter gate from activating. A naive single-threshold rule is therefore ruled out. Any concrete design must provide:

- **Hysteresis** — distinct activation and deactivation thresholds (higher to turn ON, lower to turn OFF).
- **Persistence** — the condition holds across multiple consecutive windows before it counts.
- **Delayed activation** — detection schedules activation at a known later height; it never switches immediately.
- **No oscillation** — the gate cannot flip ON/OFF within short horizons.
- **Incentive compatibility** — a miner must never gain by deliberately refusing NODE_BIND.

Prerequisite: node UX

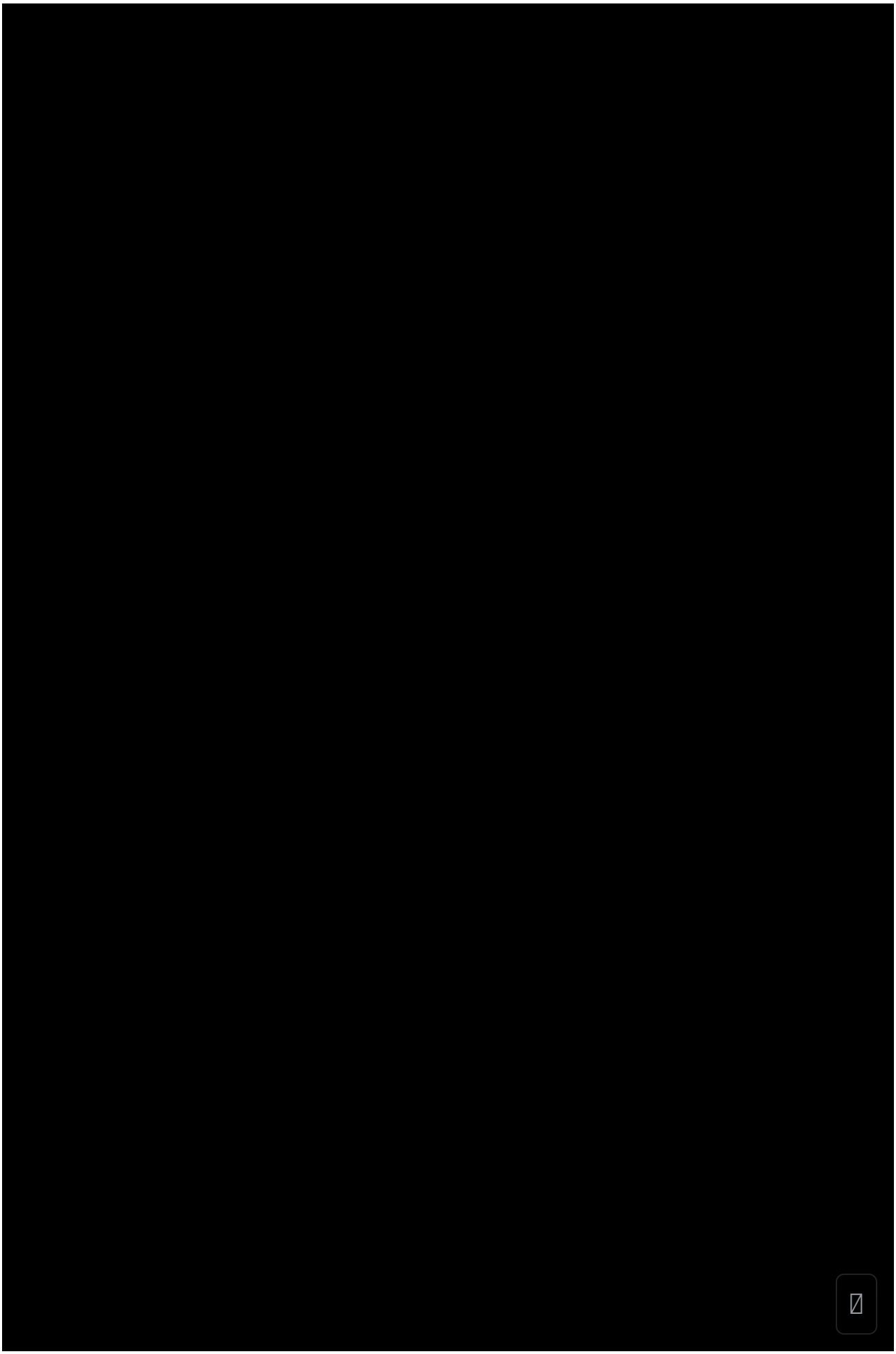
Before the requirement is extended, running a node should be close to one action for users (design direction, not a shipped feature): run node · run miner · create NODE_BIND · automatic heartbeats · Jackpot eligibility.

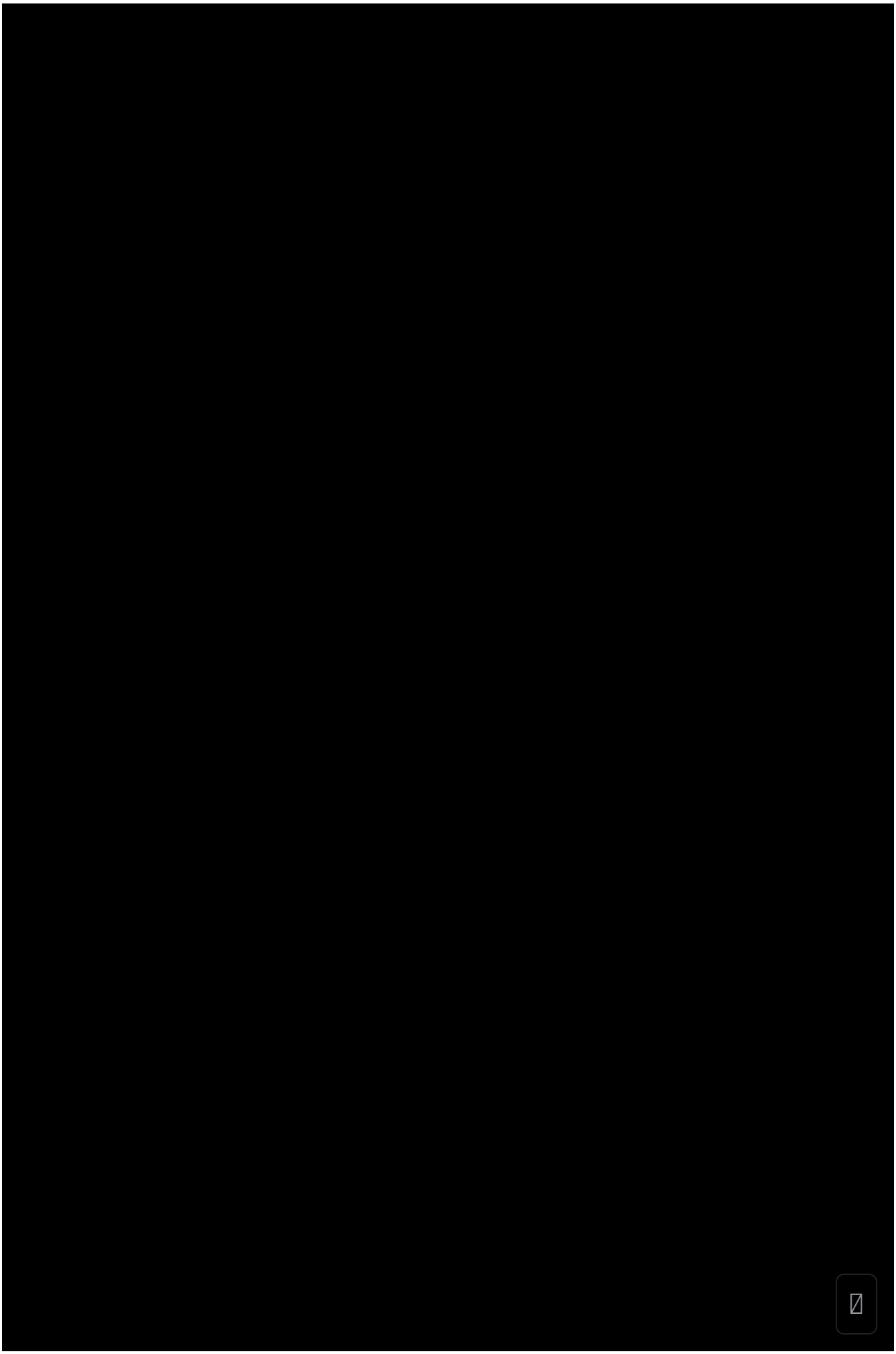
Path to consensus

Activation would require, in order: an implementation; simulation of the maturity metric and of the incentive effects; tests; review and external audit; a public notice; and an explicit, height-gated activation rule. This proposal is **not** tied to the planned #40,000 reward split (80% miner / 15% DTD normal / 5% DTD Jackpot); the two are separate decisions.

Terminology. NODE_BIND binds a mining key to a node key with two signatures. It proves a cryptographic link — a bound node identity, verified node participation — not the uniqueness of a physical machine.

Current operator documentation: **upgrade guide** · quick start · protocol spec.

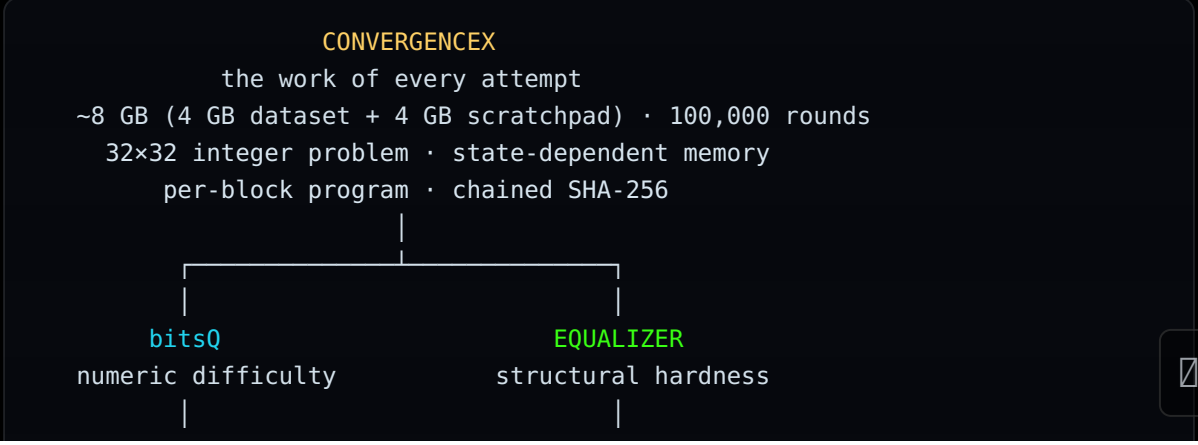


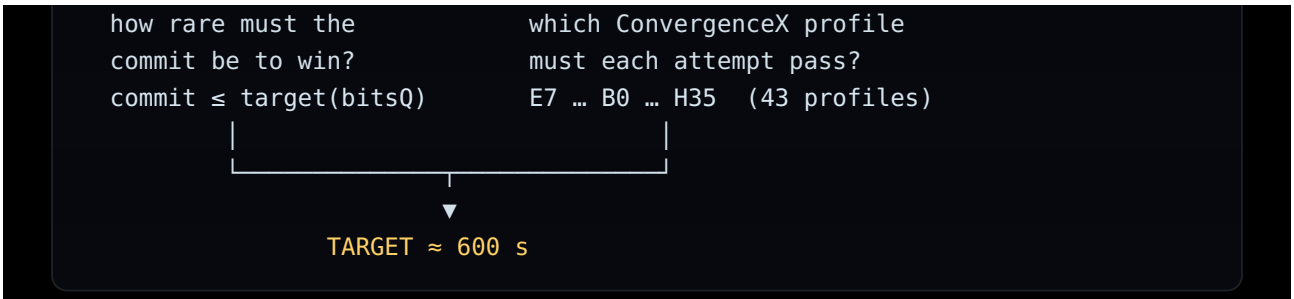


CURRENT MAINNET RULE · VERIFIED AGAINST THE NODE SOURCE

One Proof of Work, two dials: ConvergenceX + bitsQ + Equalizer

SOST keeps blocks at about **600 seconds** with two independent controls acting on the same Proof of Work. One decides **how rare a winning result must be**; the other decides **how demanding each attempt is**. Every node recomputes both and rejects a block that declares anything else.





CONVERGENCEX

Decides **what work** a miner must run: about 8 GB of memory, 100,000 sequential rounds over a 32×32 integer problem, memory reads that depend on the running state, a program generated for each block, and a stability check. The result is a 32-byte **commit**.

bitsQ

Decides **how exceptional** the commit must be. A block is valid only if commit ≤ target(bitsQ). Blocks too fast → bitsQ up, target down, harder. Too slow → bitsQ down, target up, easier. Q16.16 fixed point, integer-only.

EQUALIZER

Does **not** touch the target. It selects one of 43 profiles (E7 easing … B0 normal … H35 maximum hardening) that sets the stability parameters every ConvergenceX attempt must satisfy.

THE IDEA THAT MATTERS

SOST keeps two different things apart: the **difficulty of hitting** a winning result (bitsQ) and the **cost of making one attempt** (Equalizer). Bitcoin has only the first: one SHA-256d attempt always costs the same, and only the target moves. This does not make SOST "better than Bitcoin" by itself; it is a richer control architecture for a young network.

BITSQ — THE EXACT CURRENT RULE (BLOCK 5,270+)

```
avg = average of the last 288 block intervals        (each clamped to 1 s … 1 day)
dev = avg - 600 s

|dev| ≤ 15 s        no change                dead band
|dev| ≤ 60 s        at most 0.5 % per block
|dev| ≤ 120 s       at most 1 %
|dev| ≤ 240 s       at most 2 %
```

```
|dev| > 240 s       at most ≈3.03 %   (prev / 33)
inside the cap the step is proportional:  prev × (|dev| - 15) / 2400

Slingshot (block 7,350+), same block, if the previous block was slow:
  >20 min -6.5 % · >30 min -12.5 % · >1 h -25 % · >2 h -37.5 % · >3 h -50 %
```

EQUALIZER — THE EXACT CURRENT RULE (DIRECT LAG MAP, BLOCK 5,323+)

```
lag = (h - 1) - floor((prev.time - genesis) / 600)      positive = chain AHEAD of schedule

lag ≤ 0   →  at most B0          never harden when behind schedule
lag > 0   →  H(lag), up to H35   up immediately; down by at most max(1, elapsed/600) levels pe

Easing profiles (E1…E7) come only from:
  V12 cascade (block 7,350+): this block already took ≥ 540 s
      drop 1 · 3 · 6 · 10 · 15 · 21 · 28 levels at 540 · 600 · 660 · 720 · 780 · 840 · ≥900 s
  anti-stall: no block for 60 min → step down toward B0, then toward E7 after 6 more hours

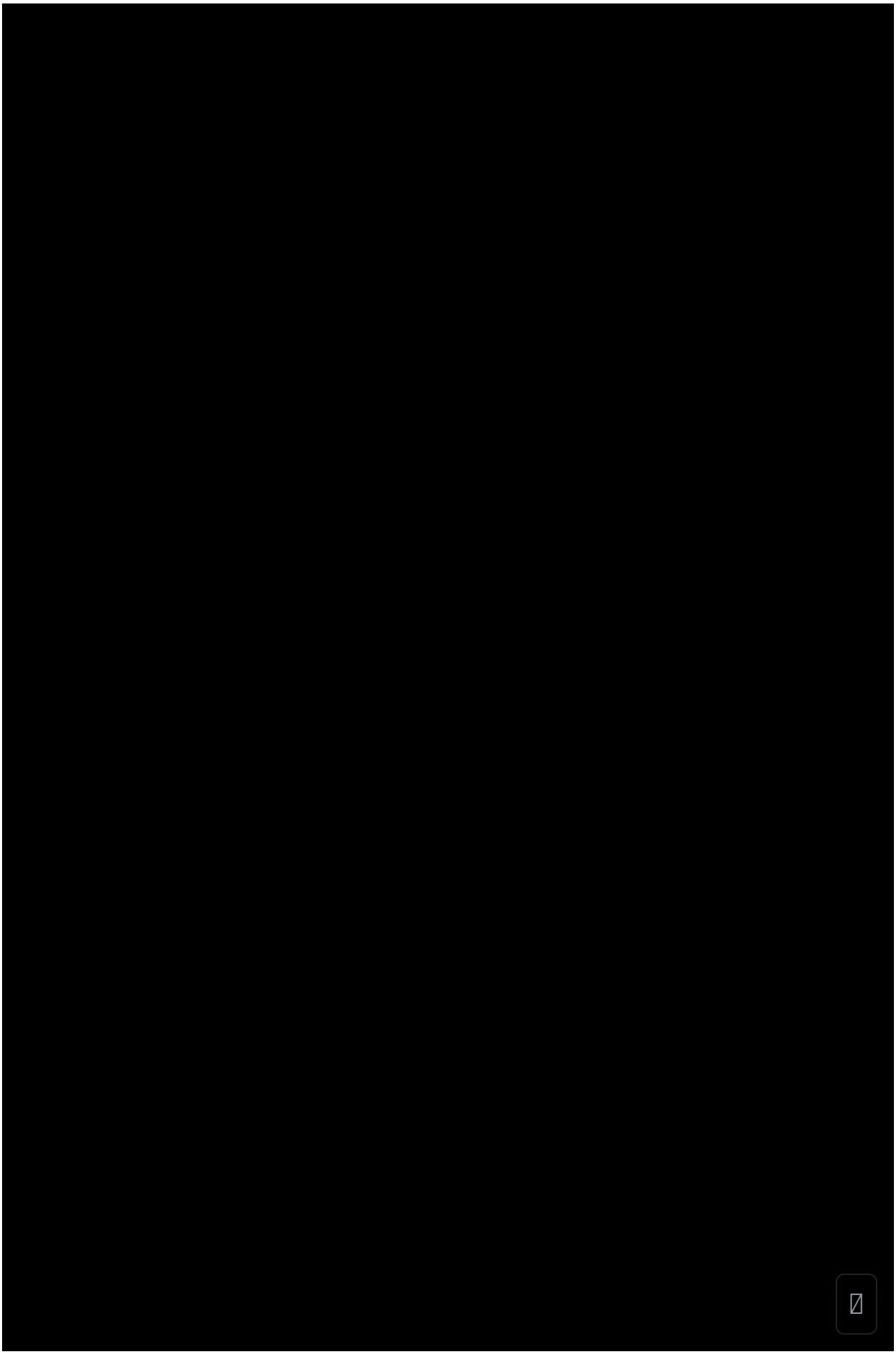
The Equalizer changes only stab_scale / stab_steps / stab_k / stab_margin — never bitsQ.
Validators recompute the profile; the block's declared profile must match exactly.
```

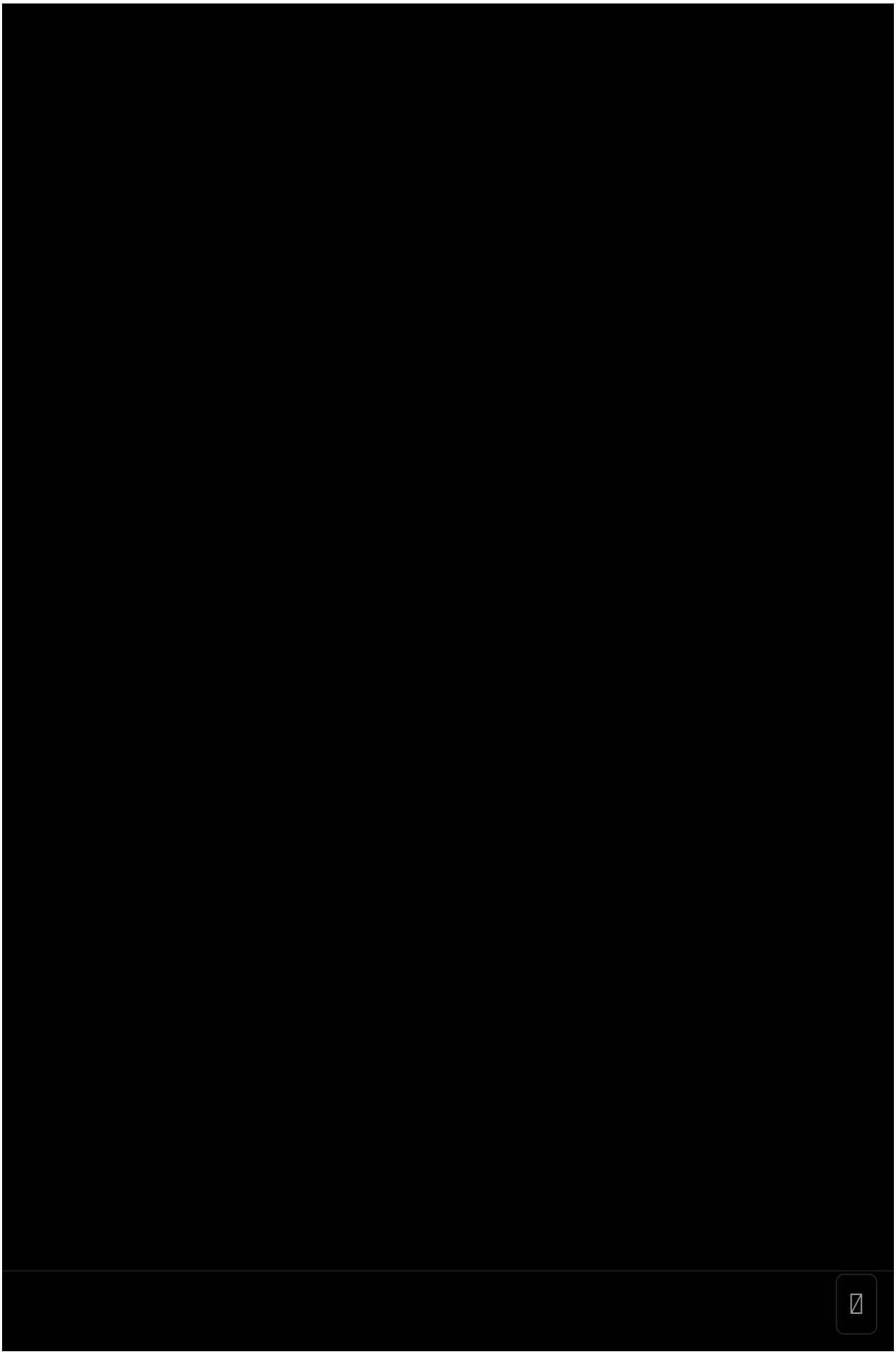
WORKED EXAMPLE

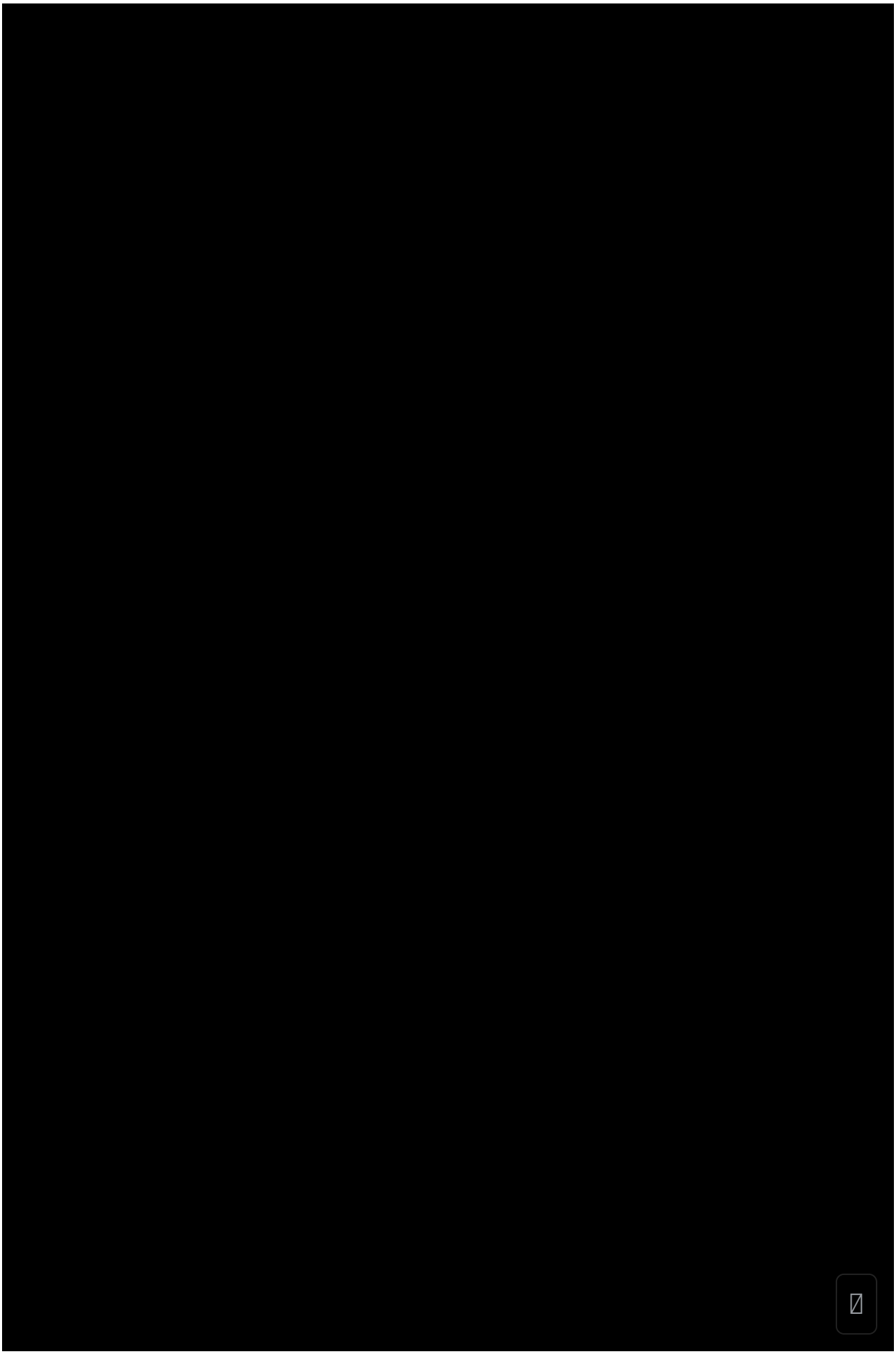
1. The chain is 4 blocks ahead of the 600 s schedule → lag = 4 → profile H4
   Each attempt must pass stricter stability parameters. bitsQ is unchanged.
2. The next block has already taken 11 minutes (660 s) → cascade drops 6 levels
   H4 → E2 for this block only: the attempt becomes cheaper, the chain catches up.
3. The last 288 blocks averaged 640 s → dev = +40 s
   proportional step = 25/2400 ≈ 1.04 %, capped at 0.5 % → target slightly easier.

> **IN SIMPLE TERMS**
>
> ConvergenceX is the exam. bitsQ is the pass mark. The Equalizer is how long and hard each exam paper is. If students finish too fast, SOST can raise the pass mark (bitsQ) and, when the class is clearly ahead of schedule, hand out a harder paper (Equalizer). If they are late, the paper gets easier first, and the pass mark follows the 288-block average.

Historical designs — the five-signal PID/EWMA equalizer used before block 5,323, the H13/H20 ceilings of earlier versions, and the pre-genesis "L1–L6" level concept — are **not** current rules. They are kept for reference in the historical appendix only.







// CODEBASE FOOTPRINT

Codebase vs Bitcoin Core

Lines-of-code totals for the SOST stack as of **2026-05-10**, side by side with the Bitcoin Core reference implementation. LOC is a rough engineering metric — **not** a quality, security or decentralisation metric — but it gives a sense of how much surface area the project covers.

Component	Σ SOST	₿ Bitcoin Core
Blockchain core (C++ / protocol)	36,740	~280–310k
Consensus & integration tests (C++ / Python)	37,187	—
Front-end & explorer (HTML + JS + CSS)	76,283	—
Protocol scripts, tooling & services (Python)	24,457	—
Documentation (markdown)	37,650	(separate)
Total code (excl. docs)	**174,667**	**~280–310k**

Component	Σ SOST	₿ Bitcoin Core
Total code + docs	212,317	—

1. The blockchain core alone is 5-7× smaller than Bitcoin Core. Bitcoin Core has 15+ years of P2P hardening, scripting (Script + Taproot), thousands of tests, and dozens of network conditions handled in production. SOST does not, and pretending otherwise would be dishonest.

2. The comparison is not apples-to-apples. The SOST total includes a hand-rolled front-end and block explorer plus protocol tooling and test suites that Bitcoin Core does not contain. The comparable blockchain core is the 36,740-line figure above.

3. What the numbers do show. SOST is a serious open-source project (~212,317 LOC including documentation) with depth beyond a typical altcoin fork — the consensus is custom (ConvergenceX + cASERT), the wallet model is in-house, the front-end is hand-rolled, and SOST is integrated with two external scientific systems.

4. What the numbers do NOT show. Network effect, peer count, security audits, deployed value, decentralisation. Those are independent of LOC and Bitcoin leads on every one of them.

Counts measured 2026-05-10 with find ... | xargs wc -l over each component's own source tree, excluding build artefacts and vendored code. Bitcoin Core figure is the upstream bitcoin/bitcoin master snapshot, code + tests, no docs.

// TRADING

DEX and OTC P2P — simple & secure

SOST ships two trading surfaces. Both are **non-custodial**, both are **simple by design**, and both have an explicit security model. Fiat trades are **not offered as a consensus feature** — the SOST chain cannot verify a bank, so SOST refuses to pretend it can.

PoPC is a native bond — not a traded product

Proof of Personal Custody (PoPC) is a single non-tradeable native SOST bond, not a market instrument. In V15 there is **no PoPC DEX** and no PoPC

contract / position trading: a bond is created, locked, and claimed by its owner, never bought or sold on an order book. The native SOST bond is the only collateral and the only slashable asset; gold is an optional reward boost only — never collateral, never escrowed, never slashed. The real trading surface is the **Atomic Swap DEX / OTC** described below.

⚠ HISTORICAL — SUPERSEDED DESIGN

The subsection below describes an earlier "SOST DEX (PoPC)" that traded PoPC contracts collateralised by tokenised gold held in an Ethereum SOSTEscrow.sol contract (the old Model B). **This is historical.** PoPC is now a single native SOST bond with no contract trading and no gold escrow; gold is an optional boost only. The text is kept for reference.

SOST DEX (PoPC) — historical design

The SOST DEX traded PoPC contracts (Proof of Personal Custody positions backed by tokenised or physical gold). Every trade combined three independent security layers:

- **Client-side ED25519-signed offers.** The offer payload (action, position-id, maker SOST address, maker ETH address, SOST amount, gold amount, price, expiry) is signed in the user's browser. The signing key is encrypted with Argon2id from the user's passphrase and never leaves the device.
- **End-to-end encrypted relay.** X25519 envelopes carry the signed offers between users. The relay server transports bytes but cannot read the cleartext, cannot alter the payload (MAC verified client-side), and cannot impersonate either party.
- **On-chain Ethereum escrow.** Where gold tokens move (XAUT, PAXG), they sit in the experimental SOSTEscrow.sol contract. **SOST holds no keys to that contract.** Releases happen by on-chain logic the two parties trigger themselves — no admin, no operator, no oracle.

SOST cannot reverse trades, cannot recover funds sent to a wrong address, cannot compel delivery, and cannot verify off-chain promises. **That was the design.** What it did instead: enforced structural integrity of every signed payload, anchored every trade in cryptographic primitives, and surfaced 26 known scam patterns via the client-side Sentinel chat layer.

SOST OTC / P2P Board — crypto-to-crypto via atomic swap

The community OTC / P2P Board hosts user-to-user trades against seven counterparty assets: **BTC, ETH, USDT, USDC, BNB, PAXG, XAUT**. The intended settlement mechanism is the **atomic swap HTLC** (Hashed Time-Locked Contract): cryptography enforces that **either both parties receive their funds, or both refund** — no intermediate state exists in which one side walks away with everything. No SOST escrow, no admin custody, no first-mover risk, no protocol authority.

The implementation status (as of this revision) is:

- **SOST-side consensus** — **LIVE since #16,000 (V14.5); relay/policy flag-day #17,000 (V14.7)**. HTLC_LOCK (R17), HTLC_CLAIM (R19-R22), HTLC_REFUND (R23-R24) validation rules are in src/tx_validation.cpp; nodes have accepted HTLC transactions since #16,000.
- **Wallet, RPC, and CLI** — **complete**. createhtlclock, claimhtlc, refundhtlc, decodehtlc, gethtlcstatus are available.
- **EVM counterparty leg** — **lab-verified** (AtomicSwapHTLC.sol, 57 Foundry tests + full end-to-end run on a local Anvil node); **not deployed on mainnet.**
- **BTC counterparty leg** — **lab-verified** on Bitcoin Core regtest (P2WSH funding, preimage claim, refund, reorg test); BTC signing is **OFF by default** (CMake flag) and **not compiled into the official binaries**.
- **Public use** — real cross-chain swaps remain **founder-testing only**, pending an external cryptographic + economic review. The community OTC board can coordinate swaps, but the protocol provides no escrow, custody or dispute resolution.

The trust profile of the seven assets splits into two categories. **Trust-minimized (BTC, ETH, BNB)**: no token-issuer freeze risk; cryptographic atomicity holds on both sides. **Issuer-risk (USDT, USDC, PAXG, XAUT)**: the underlying token can be frozen by its issuer (Tether, Circle, Paxos, TG Commodities); if that happens mid-swap, the SOST side still refunds correctly but the counterparty side may become uncollectible. The DEX UI labels these four explicitly and recommends small amounts for them; larger amounts should stick to Category A.

OTC against fiat — deliberately not offered as a consensus feature

A trade where one leg is a SEPA transfer, wire, or cash payment **cannot be made cryptographically atomic.** The SOST chain has no view into a bank. Any mechanism that would "penalise the unpaid fiat side automatically" requires either (a) an oracle that reads the bank —

which is arbitration and turns SOST into a regulated intermediary — or (b) one party self-declaring incumplimiento, which is a trivially-exploitable robbery primitive; both also enable Sybil collusion farms. There is no third path.

Therefore **SOST does not offer fiat OTC as a consensus service.** The community discussion board still permits conversations about fiat trades (people are free to do whatever they want with each other), but SOST provides no escrow, no penalty, no dispute resolution, and no implied safety. **SOST's strong recommendation is to keep all trades in the crypto-to-crypto path** (atomic swap) where the cryptography itself can enforce safety. Any fiat trade is off-protocol, at user risk, with no protocol safety net.

The simplicity rule

Both surfaces follow the same five-step UI discipline so the user can be safe without becoming a cryptographer:

1. Load only the official URL: sostcore.com (the only official domain; the former sostprotocol.com is being retired). Never click a link from a DM.
2. Read the signed offer fields before signing. Counterparty address, amount, expiry — all visible, all binding.
3. Verify counterparty addresses against the LOCK / escrow payload. The signed payload is authoritative; chat messages are not.
4. Confirm txids on-chain (SOST explorer or Etherscan) before releasing anything. Screenshots are not proof.
5. Do a small test transaction before any larger amount. Always. Without exception.

Anything sold outside this flow — Telegram DMs, "official escrow admin", address changes by chat, advance fees, screenshot proofs — is a scam pattern. The client-side Sentinel flags 26 known variants at the chat layer. The OTC and DEX pages document each pattern with worked examples. **If a counterparty pushes you off the official flow, stop the trade.**

// AUTONOMOUS CHAIN SAFETY · updated 2026-09-29

Autonomous Chain Safety (SACS)

SOST is an open Proof-of-Work chain: every node independently verifies blocks and follows the chain with the **highest cumulative verified work** (not the longest, and never a peer's self-reported work). SOST does **not** use ChainLocks, masternodes, validator quorums, miner votes or a developer-selected chain. It therefore offers **probabilistic PoW security**, not mathematically irreversible finality: a sufficiently powerful adversary that produces a valid higher-work chain cannot be excluded by software alone.

SACS (SOST Autonomous Chain Safety) is a safety layer with two clearly separated parts:

- **Current mainnet.** Best-chain-by-cumulative-work selection; atomic reorg with full rollback; **MAX_REORG_DEPTH = 500 blocks (~3.5 days)** — a reorg-depth rule enforced in code: a reorg at/below the cap converges to the most-work chain, a reorg deeper than the cap is rejected (the node keeps its chain even against more valid work — confirmed in a scaled devnet lab), which bounds deep reorgs but can prevent automatic convergence after a very long partition; hard checkpoints (≤ height 3,554) reject a contradicting chain; 1,000-block coinbase maturity. The 500-block rule bounds deep reorgs but, after a long partition, can prevent automatic convergence to a valid higher-work chain (confirmed in a scaled devnet lab: at/below the cap converges, above it splits persistently). Resolving that safely — treating the depth as an alert plus full re-verification — is the devnet-only deep-reorg research, post-fork.
- **SACS Core — LAB VERIFIED (V16-compatible, not yet deployed).** Fork/reorg monitoring, structured chain-safety events, a read-only getchainsafety RPC, a transaction-safety state machine (CONFIRMED → REORGED → REENTERED_MEMPOOL → CONFLICTED) and reconciliation for the wallet, Explorer, DEX and exchanges, plus recovery/persistence hardening. This is observability and application protection — **it changes no consensus rule**: a SACS alert can make an application wait

or pause, never turn a valid block invalid, never halt honest mining. Blockchain truth outranks any local database or UI.

- **SACS Deep-Reorg Recovery — FUTURE RESEARCH (devnet only).** Studies treating the 500-block depth as an **operational alert** rather than an automatic rejection, so that after a long partition a node can fully re-verify the alternative branch (work, UTXO, emission, difficulty), protect sensitive operations, and converge on the valid higher-work chain. Any such change is consensus-affecting and would require a future coordinated upgrade — it is **not** part of the V16 fork at block #30,000.
  Status: SACS Core is code-complete and lab-verified on a research branch; it is **not** live on mainnet and no node binary is swapped before #30,000. SACS provides autonomous detection, recovery support and transaction-safety signals — it does not provide absolute, irreversible finality.

⬢ MOST LIKELY CURRENT RESEARCH SCENARIO — NOT CONSENSUS

The current leading scenario under study is a future activation around **block #40,000 or #50,000**, subject to network maturity, miner distribution and further simulation. **80/15/5 supersedes 75/20/5 as the current leading research scenario.**

80%	15%	5%
direct PoW miner reward	normal DTD per block	into the existing DTD Jackpot pool

80% — direct PoW miner reward. Miners contribute measurable hashpower, valid block production and computational/economic chain security.
15% — normal DTD distribution continues per block.
5% — into the existing DTD Jackpot pool. This 5% **accumulates in the DTD Jackpot pool that V30000 already distributes** — it is **NOT a direct 5% node payment per block.** When jackpots are paid, the existing V30000 DTD Jackpot eligibility framework remains applicable, **including the node / NODE_BIND eligibility requirements.**

This gives **two complementary incentives:** (1) a **direct PoW incentive** for the miners securing the chain, and (2) an **accumulated Jackpot incentive** for eligible network participants that maintain the required network infrastructure.

80% → direct PoW miner reward (chain security)
15% → continuous normal DTD per block
5% → accumulates in the existing DTD Jackpot pool (not a direct node payment)

No new supply. All three allocations come from the **existing block emission. Not consensus.** Blocks #40,000 and #50,000 are the two most likely activation scenarios — **neither height is final.** The **80/15/5** structure is the **leading research candidate, not active consensus** (current consensus is unchanged: 50% miner / 50% DTD) and supersedes 75/20/5 as the leading scenario. Any implementation would require explicit consensus rules and a **future hard fork. SOST is also actively pursuing full decentralization of the system as soon as possible.**

// DECENTRALIZATION

Honest Assessment

The network and mining layers are already partially decentralized: multiple independent nodes, multiple active miners, open-source code, and no admin keys for block production.

However, two economic custody components are still temporarily centralized during Phase I (until governance activates — originally targeted for V6 / block 10,000, deferred to **V14 / block 15,000** after the 2026-05-18 scope review; see banner above and docs/V13_PUBLIC_SCOPE_UPDATE.md):

- **Gold Vault** (25% of every block reward (until #25,000; 0 since)): the 5-defense governance model once targeted for V14 / block 15,000 was **deferred and never activated** and is now **paused indefinitely / legacy**. From **V15 (block 25,000)** the accumulated reserve is redistributed on-chain to miners via the **DTD / DTD Jackpot** architecture (supply-neutral). Historical audit at docs/V13_GOLD_VAULT_GOVERNANCE_GATES.md.
- **PoPC Pool** (25% of every block reward **until block #25,000 — 0% since**): developer-held custody of what it already received. The single-model PoPC bond (native SOST bond only — gold is an optional reward boost, never PoPC collateral, never escrow, never slashed) is **not operational.** **Correction:** an earlier draft of this paragraph said the DTD reward would require an active PoPC bond from block 30,000.

It does not. On mainnet the PoPC eligibility gate ships deactivated (DTD_POPC_GATE_CONSENSUS_ACTIVE = false): **the DTD is permissionless — mining one valid block is the only requirement**, and V16 at #30,000 changes which miners qualify, not whether a bond is needed. The Gold Vault / Metals Reserve (the other 25%) is a separate channel and is **not** the PoPC reward source. Full audit at docs/V13_POPC_ESCROW_AUTO_ACTIVATION_GAPS.md

SOST developer cannot change the constitutional 50/25/25 split (blocks 0–24,999; 50/50 miner/DTD since #25,000), because it is hardcoded at genesis. What remains centralized during Phase I is custody, not the emission rule itself.

Custody transition roadmap:

- **Block 15,000 (V14 hard fork):** H3/H4 consensus hardening. The **Gold Vault governance** model (5-defense: purpose restriction, dual destination whitelists, per-spend cap, rate limiting, ≥90 % miner signaling over a 67-block window, plus the Transitional Guardian role) and the **PoPC single-bond** once proposed for enforcement here / at V15 were **deferred and never activated**. Both are now **paused indefinitely / legacy**, superseded by the **DTD / DTD Jackpot** architecture: from V15 (block 25,000) the accumulated Gold Vault + PoPC reserve is redistributed to miners on-chain (supply-neutral), and from V16 (block 30,000) the DTD Accumulated Reward V2 becomes an independent node-gated draw. Governance threshold history retained for reference: 75 % (original whitepaper) → 95 % (V6 internal review) → 90 % (V13 review).
- Gold reserve purchases become externally verifiable on Ethereum (already started: 1.2 oz committed)
- **Phase III (12–24 months):** Migration from tokenized gold (XAUT/PAXG) to physical gold custody in a regulated vault. No bridges, no wSOST, no cross-chain attack surface.

The custody arrangement during Phase I is at SOST developer's operational discretion. The transition to network custody at block 10,000 has a hard date encoded in the consensus rules — it is not subject to the developer's later willingness to step back. The Gold Vault must be able to change form, but not purpose.

SECURITY MODELS · SIMPLE COMPARISON WITH EXAMPLES

SOST vs Bitcoin vs Ethereum: who secures the chain, and how are they paid?

Two different questions are mixed up all the time: **how a protocol is designed**, and **how decentralized its real network is**. Bitcoin and Ethereum are remarkable not only for their design but because thousands of independent people and organisations run them, verify them and have real money at stake, year after year. SOST shares the design principles of permissionless consensus; it does not yet have that maturity.

BITCOIN · WORK

SHA-256d Proof of Work on specialised ASICs. A block about every 10 minutes; difficulty retargets every 2,016 blocks. The block finder (in practice usually a pool) takes the whole subsidy — 3.125 BTC since the 2024 halving — plus fees. Full nodes are not paid, but each one verifies every rule on its own.

```
ASIC + electricity → SHA-256d attempts
→ hash < target → block → BTC
```

ETHEREUM · STAKE

Proof of Stake since 2022. A solo validator deposits **32 ETH**. Time runs in 12-second slots; 32 slots = one epoch (6.4 min). Validators vote (attest) about once per epoch and are occasionally chosen to propose a block. Rewards come from issuance plus, for proposers, priority fees and MEV; the base fee is burned. Equivocation is slashed.

```
32 ETH deposit → validator
→ votes + occasional block → ETH
```

SOST · WORK + PARTICIPATION

ConvergenceX Proof of Work on ordinary CPUs and RAM. A block about every 10 minutes; bitsQ and the Equalizer adjust every block. The block reward is split 50% to the finder and 50% to the DTD draw among recent miners. A separate Jackpot every 288 blocks rewards miners who also run a bound node.

```
CPU + RAM + electricity → ConvergenceX
→ commit ≤ target → block → 50% + DTD
```

THE SAME DAY, THREE WAYS — WORKED EXAMPLES

BITCOIN A miner with 1 % of the network hashrate finds on average 1 % of ~144 daily blocks ≈ 1.4 blocks/day, each worth 3.125 BTC + fees. Nothing else. Running a node earns nothing, but it lets you reject any block that breaks the rules — even one from the largest pool.

ETHEREUM A validator with 32 ETH attests roughly once every 6.4 minutes and, now and then, is selected to propose a block (bigger reward). The yearly rate is a few percent and moves with the total amount staked. Offline: small penalties, not slashing. Signing two conflicting blocks or votes: slashed — part of the 32 ETH is destroyed.

SOST Epoch-0 subsidy is 7.85100863 SOST per block. Miner A finds a block with no fees:

A, the finder, receives	3.92550432 SOST	(odd stock goes to the miner)
DTD half	3.92550431 SOST	

The DTD half goes to ONE eligible recent miner drawn uniformly — it may be A, or it may be B, a much smaller miner.
Every 288 blocks the DTD Jackpot (100 SOST base, up to 500 with rollover, paid from the frozen historical reserve, no new coins) goes to a miner who has ≥ 3 PoW blocks in the last 2,016, an active NODE_BIND and the heartbeat history; the draw is weighted by PoW blocks.

SIDE BY SIDE

	Bitcoin	Ethereum	SOST
Security	Proof of Work (SHA-256d)	Proof of Stake	Proof of Work (ConvergenceX)
What you commit	ASICs + energy	32 ETH (solo validator)	CPU + RAM + energy; no deposit
Block producer			

	whoever finds the PoW	validator selected per slot	whoever finds the PoW
Votes / attestations	no	yes	no
Slashing	no	yes	no
Direct reward	100% subsidy + fees to the finder	issuance; proposer also gets fees/MEV	50% of subsidy + fees to the finder
Extra reward	—	occasional block proposal	DTD every block · Jackpot every 288 blocks
Are nodes paid?	no	only as validators	no — a bound node is a Jackpot condition, never a multiplier
Sybil resistance	hashrate	staked ETH	real PoW blocks (SbPoW-signed identities)
Coins locked to take part	no	yes	no
Finality	probabilistic (more blocks = safer)	explicit, after ~2 epochs (~13 min)	probabilistic
Difficulty adjustment	every 2,016 blocks	n/a	every block: bitsQ + Equalizer
Long-term security budget	fees after the last halving	issuance + fees	fees only after the cap (no tail emission); under current rules half of fees also go to DTD

WHY SOST DOES NOT PAY NODES SIMPLY FOR EXISTING

Paying every node invites a Sybil attack: one person rents 10,000 cheap servers and claims 10,000 rewards. The Internet cannot tell 100

IPs that are 100 people from 100 IPs that are one person. Jackpot V2 therefore uses the node only as a **door**, and real PoW as the **weight**:

```
1 miner · 100 nodes · 100 PoW blocks   → weight 100
1 miner ·   1 node  · 100 PoW blocks   → weight 100
1 miner ·   5 nodes ·   2 PoW blocks   → not eligible (minimum 3)
Creating nodes adds nothing. Doing work does.
```

Normal DTD today: an address must have mined in the last 288 blocks; addresses that mined any of the previous 6 blocks are skipped (with a fallback if that would leave nobody); the draw is uniform among the eligible, not proportional to hashrate. An anti-dominance rule (an address with ≥10% of the last 288 blocks is skipped) exists but switches on only when at least 11 distinct miners are active — so with today's small miner set it is not in effect. Jackpot V2 has no cooldown and no anti-dominance.

THE HONEST TEST OF DECENTRALIZATION: WHAT CAN YOU SWITCH OFF?

BITCOIN

Switch off bitcoin.org, the largest exchanges, the largest pool, GitHub and a thousand nodes: blocks keep coming. That is the standard.

ETHEREUM

Switch off ethereum.org, the Foundation, a large RPC provider, a major staking service or one client implementation: with enough remaining validators, the chain continues.

SOST TODAY

The project still provides key infrastructure (seed nodes, the public explorer and RPC, release distribution), only a few independent operators exist, and two addresses produced the last 288 blocks. The goal is that switching off everything run by SOSTCore changes nothing for the chain. We are not there yet.

Property	Bitcoin	Ethereum	SOST (goal, if reached)
Permissionless consensus	yes	yes	yes (today too)

Your own node verifies all rules	yes	yes	yes (today too)
Home hardware competitive in block production	no (ASICs)	node yes; validating needs 32 ETH	the design goal — not yet proven under adversaries
Client diversity	moderate	very high	one implementation today
Independent operators	enormous	enormous	must grow a lot
Years under attack	~17	~11	months
Dependence on the creator	none	very low	goal: none

IN ONE SENTENCE EACH

Bitcoin is minimalist decentralized money: it does few things and does them with extraordinary resistance. **Ethereum** is decentralized computation: it accepts much more complexity to run programs, tokens and markets. **SOST** aims to be Proof-of-Work money on a Bitcoin-like base (UTXO, 10-minute blocks), with ConvergenceX for broad CPU mining, per-block difficulty control, and DTD incentives that reward work and node participation — without turning into a staking chain. There is no "lock SOST and earn" product, by design.

WHAT CANNOT BE WRITTEN IN C++

ConvergenceX, bitsQ and the Equalizer can build a good architecture. The final decentralization is not in the code: it comes from independent people running that code without needing us. ASIC resistance by memory hardness is a design goal that has to survive years of adversaries; it is not a proven fact.

STATUS: RESEARCH / DEVELOPMENT · NOT ACTIVE · NOT READY FOR PRODUCTION

SOST Atomic Bridge (Atomic Reserve Vault), Gold Access Key, Gold Reserve V2 and governance (#40,000 research horizon)

The target architecture of the bridge is the SOST Atomic Bridge (3.16.1). C701 replaced the previous federated design with automated verification; its lock/mint model is now the conventional reference (3.16.2). The bridge, wSOST and Gold Reserve V2 exist only as development contracts, a verification-engine prototype and tests on a **local development chain**. Nothing is deployed on any public chain, no contract is live, no funds exist and there is **no launch date**; real operations are **blocked by design** until the open blockers below are closed, and any future activation is subject to security work, an external audit and legal/regulatory review. Current consensus (50% miner / 50% DTD) is unchanged; #40,000 is a research horizon, **not an activation commitment**. Public explanation and DEMO: wSOST & Gold Vault.

TWO SERVICES ONLY

SOST Bridge	native SOST ↔ wSOST on Ethereum, 1:1 (this section)
SOST Atomic Swap	separate P2P HTLC service: no SOST↔wSOST pair, no Gold Access Key, never to

3.16.1 TARGET ARCHITECTURE — SOST ATOMIC BRIDGE (ATOMIC RESERVE VAULT)

STATUS: TARGET ARCHITECTURE · RESEARCH / DEVELOPMENT · NOT ACTIVE · NOT MAINNET · NOT AUDITED · NOT YET PROVEN

On 2026-10-26 SOST fixed one target architecture for the bridge: the SOST Atomic Bridge, an Atomic Reserve Vault (ARV). SOST does not intend to mint a new wSOST for every bridge operation. The target uses a preissued, immobilized wSOST reserve on Ethereum and bidirectional atomic swaps; native SOST received by the bridge forms the corresponding reserve. The conventional lock/mint design of 3.16.2 (C701/C703) is kept as the reference model and as the prototype whose components are reused; it is not a second product.

Premint accounting:

```
TOTAL_PREMINTED_wSOST   fixed at deployment, no minter afterwards
IMMOBILIZED_wSOST       free inventory held by the Ethereum Atomic Vault
PENDING_wSOST           inside entry / exit HTLCs of the vault
CIRCULATING_wSOST       TOTAL - IMMOBILIZED - PENDING
BACKING_NATIVE_SOST     confirmed native reserve outputs   (+ PENDING_NATIVE_SOST in HTLCs)
rule      CIRCULATING_wSOST <= CONFIRMED_NATIVE_SOST_RESERVE  (equal after every swap)
not       totalSupply(wSOST) <= reserve  (totalSupply includes the immobilized inventory)
example   premint 1,000,000 = 990,000 immobilized + 10,000 circulating; reserve 10,000 → 100%
```

Native SOST is never burned. Returned wSOST goes back to the inventory: no burn is required. A large inventory is itself a target; it would be sized to the exposure caps and topped up in tranches (the 1,000,000 figure is illustrative).

Entry (SOST → wSOST, 1,000) with the native HTLC active since #16,000 and an Ethereum HTLC with the same sha256 hashlock:

```
1 wallet creates s, H = sha256(s); request bound to H, amount, recipient; XAUT escrowed
2 user locks 1,000 SOST: OUT_HTLC_LOCK(H, claim = reserve, refund = user, +288 blocks)
3 engine waits 30 confirmations and checks the lock
4 vault moves 1,000 EXISTING wSOST into an HTLC(H, beneficiary = user, refund to vault, ~12 h)
5 user claims with s: wSOST to user, escrowed XAUT to Gold Reserve V2      (no mint)
6 reserve claims the SOST with the revealed s
reserve 10,000 → 11,000 · circulating 10,000 → 11,000 · inventory 990,000 → 989,000
```

Step 4 must not happen for a SOST lock that does not exist, so the entry still needs a verifier: today the engine's authorization (a threshold of keys is possible), later possibly a SOST light client on Ethereum. Mint authority becomes release-from-inventory authority, bounded by the inventory and the caps, not removed.

Exit (wSOST → SOST, 500):

```
1 user creates s', H' = sha256(s'); locks 500 wSOST in HTLC(H', beneficiary = vault, ~48 h)
2 engine waits for Ethereum finality
3 reserve funds OUT_HTLC_LOCK(H', claim = user, refund = reserve, +72 blocks), 500 - fee
4 user claims the SOST with s'
5 anyone settles the Ethereum HTLC with s': 500 wSOST back to the inventory      (no burn)
reserve 10,000 → 9,500 · circulating 10,000 → 9,500 · inventory 990,000 → 990,500
```

Failure and refund: the side that moves first gets the longer deadline. If the vault never releases, the user refunds the SOST; if the user never claims, the vault HTLC returns to the inventory and the user refunds; the XAUT escrow returns on expiry; secrets and swap ids are single-use. Two risks stay open: (a) if the user claims wSOST and the reserve fails to claim the SOST before the user's deadline, the

user could refund and keep the wSOST (only the claim key can claim a SOST HTLC, so watchtowers cannot help); (b) a reorg of the SOST lock after the vault released is bounded only by the 30-confirmation depth. The Gold Access Key is escrowed before the claim because a secret revealed in a reverted claim would otherwise let the reserve take the SOST while the user received nothing.

The central problem — who controls the native SOST reserve? An atomic swap coordinates the exchange; it does not decide who controls the SOST afterwards. The native HTLC has single-key claim and refund keys. V17 native multisig (OUT_MULTISIG, M-of-N ⩽ 15, no script engine) may fund an HTLC and receive an HTLC claim, but cannot be an HTLC claimant or refund key, and has no timelocked recovery path. A 3-of-5 reserve reduces single-key risk; three signers can still move the whole reserve, and they decide off-chain when to fund an exit. With current and V17 primitives, arbitrary withdrawal by the reserve signers is NOT impossible. Missing primitives (research only, none authorized, consensus unchanged):

```
P1 destination-committed HTLC: claim output fixed at lock time, claimable with the secret alon
P2 HTLC with a multisig claimant / refund key
P3 reserve vault with delayed spending and a recovery path
P4 covenant: reserve spendable only into exit HTLCs refunding to the reserve, rate-limited
P5 SOST able to verify Ethereum state (light client / proof)
```

Cross-chain proofs keep a role, no longer for every exchange: the entry release, public reserve verification, settlement checks, reorg detection and recovery. ConvergenceX has a compact transcript proof (about 10-20 ms natively, no 4 GB dataset), so a SOST light client on Ethereum is not ruled out; its EVM cost with the cASERT rules is unmeasured and expected to be high.

Comparison (official documentation, 2026-10): Atomiq verifies Bitcoin PoW with an on-chain light client and releases LP liquidity from UTXO-controlled vaults; THORChain uses a bonded validator set, TSS vaults and liquidity pools; Chainflip uses 150 validators and an aggregate key with FROST 100-of-150 signing. The SOST design aims to avoid a separate validator network and a pool; it is not claimed to be more secure, more advanced or novel. Neither model A (lock/mint) nor model B (ARV) is automatically trustless.

Proof of concept (testnet / devnet only, NOT STARTED): 100 fictitious preminted wSOST; entry, exit, counterparty disappears, deadline

ordering, engine stops, secret replay, concurrent swaps, reserve insufficient, reorg, attempted reserve theft; decisive test 500 SOST → 500 wSOST → 500 SOST with 1:1 kept and no operator permission. The theft test is expected to fail with today's primitives. Until all of it is shown, the Atomic Reserve Vault is RESEARCH ONLY.

C705 FEASIBILITY RESULT — A KEYLESS NATIVE RESERVE

A reserve that nobody holds a key to can only react to events its own chain can verify. Hashlocks alone cannot protect it: with no deciding counterparty, anyone could make it lock SOST under a secret of their own choosing. The keyless form of the Atomic Bridge therefore replaces "who signs" with "what is proven", in both directions: the Ethereum vault releases inventory only against a proof of a confirmed SOST deposit, and SOST pays from the reserve only against a proof of a finalized Ethereum return. HTLCs remain only as an optional fast path for market makers swapping their own funds. Minimum new SOST primitives (research only, none authorized, consensus unchanged):

```
N1 = P4  keyless ARV outputs, no pubkey, no script, fixed spend paths:
         ARV_DEPOSIT (names 0x recipient, deadline, refund address, vault)
           → swept into the reserve on a proof of EntryReleased, or
           → refunded to its fixed address on a proof of EntryCancelled
         ARV_RESERVE singleton (chained output; holds the Ethereum light-client state)
           → pays proven ExitRequested events, sequential nonce, fixed destination
N2 = P5  ETH_FINALITY_PROOF: sync-committee light client in consensus (BLS12-381, SSZ,
         keccak, RLP, Merkle-Patricia receipt proofs), finalized headers only
P1, P2, P3  not needed (no key to fix, share or delay); multisig is not used for the reserve
Ethereum  immutable vault, no admin; SOST header relay (optimistic, bonded, fraud proof by
          one ConvergenceX transcript check; gas unmeasured); each entry RELEASED before its
          deadline xor CANCELLED after it
```

Every action is permissionless: amount and destination are fixed by the deposit or the Ethereum event, so execution is open and ownership does not exist. A bounded model check (research/arv/arv_model.py) explores every order of events with users, relayers and an attacker who holds funds: a key or M-of-N reserve can be taken (DESIGN FAIL); the keyless rules lose or duplicate nothing and every user can finish or cancel alone; a SOST reorg deeper than the confirmation depth breaks the backing. Evidence, not a formal proof. Remaining trust: SOST PoW majority (weak while mining is concentrated, so amounts must be capped), more than 2/3 of Ethereum's 512-member sync committee, code correctness, one honest relayer, bootstrap checkpoints; every Ethereum hard fork needs a SOST release first, or the reserve waits

frozen. Both rules are a hard fork: NOT in V17 (#40,000); better as V18, after a devnet PoC of N1 + N2 and an external review. No bridge is 100% trustless; the aim is no custodian, no admin key, no multisig and no human approval.

3.16.2 CONVENTIONAL REFERENCE MODEL — LOCK / MINT (C701/C703 PROTOTYPE)

INVARIANTS AND BACKING (CONVENTIONAL REFERENCE)

```
Example        1,000 SOST locked in the Bridge Vault  →  1,000 wSOST minted on Ethereum
Native SOST    never burned: it is locked, and released when wSOST is burned (only wSOST is ev
Backing        backing of wSOST = native SOST locked = wSOST minted (in base units)
The XAUT contribution never reduces the amount minted
Burning N wSOST entitles the release of N SOST (minus the published SOST network fee)
No wSOST without a verified, confirmed native deposit · gold never mints wSOST · DTD Jackpot f
```

The XAUT contribution (Gold Access Key) is **NOT backing** of SOST or wSOST, **NOT a peg**, **NOT a price floor**, **NOT a price guarantee**, **NOT a redemption promise** and **NOT 1:1 gold backing**. Gold held by Gold Reserve V2 gives **no individual redemption right**.

FIVE ACCOUNTS, NEVER COMBINED

```
1 native SOST · 2 bridge-locked SOST · 3 wSOST (= box 2) · 4 Gold Reserve V2 (XAUT; PAXG gated
5 historical Gold Vault + PoPC (frozen since #25,000, committed to the DTD Jackpot)
```

FLOW OF THE CONVENTIONAL REFERENCE PROTOTYPE (LOCK/MINT)

```
SOST → wSOST   request (requestId = hash of chainId, contract, recipient, amount, vault, salt)
               → deposit to the Bridge Vault with capsule C701:0x<requestId> on that output
               → verification engine checks vault, exact amount, capsule, 30 conf. (provisiona
               → one EIP-712 authorization (chainId, contract, recipient, amount, txid:vout, e
                 + separately signed XAUT/EUR quote
               → ONE Ethereum tx: XAUT Gold Access Key → Gold Reserve V2, exactly N wSOST → us
wSOST → SOST   any holder burns N wSOST (no XAUT) → vault releases N SOST minus the SOST netwo
               (capsule C701R:0x<releaseId>) → verified and recorded on Ethereum
A transaction hash typed by a user is a search hint, never proof.
```

GOLD ACCESS KEY

```
feeEUR  = min(250, max(0.50, bridgeValueEUR × 0.001))      0.10% · min 0.50 EUR · max 250 EUR
feeXAUT = feeEUR / verified XAUT price, rounded UP to XAUT's 6 decimals (integer math)
minimum operation 100 EUR · 100% to Gold Reserve V2 · Ethereum gas paid by the user, shown sep
no XAUT contribution on the return path (wSOST → SOST)
fixed illustrative price 1 XAUT = 3,713.13 EUR (2026-10-19, illustrative, not a quote):
  100 EUR → 0.000135 XAUT · 1,000 EUR → 0.000270 · 10,000 EUR → 0.002694
  100,000 EUR → 0.026932 · ≥ 250,000 EUR → 0.067329 (cap)
```

The cap was revised from 50 EUR to **250 EUR** on 2026-10-26: at 1,000,000 EUR the effective rate is 0.025%, against 0.005% with a 50 EUR cap. Fees of other bridges price capital, relayers and gas and are not directly comparable to this contribution.

HOW IT DIFFERS FROM MOST BRIDGES

family	who decides a deposit is real	pool	who keeps the fee
custodian / multisig	custodian or N-of-M keys	no	custodian / merchants
liquidity network	relayers / pool protocol	yes	LPs, relayers, treasury
guardian / messaging	external guardian set	no	relayers / protocol
issuer burn-and-mint	the issuer	no	- (gas only)
rollup / light client	the destination chain itself	no	- (gas only)
SOST bridge (design)	SOST's automated engine	no	nobody: 100% to Gold Reserve

Most bridges pay their fee to liquidity providers, relayers, a treasury or token holders, and many depend on an outside key set or a pool (Ronin 2022: 5 of 9 validator keys; Harmony Horizon 2022: 2 of 5 multisig keys; Wormhole 2022: a signature-verification bug minted 120,000 wrapped ETH). The SOST design has no pool (always exactly 1:1), no outside federation, no fee on the return, and its fee builds a public gold reserve. It is **not trustless**: in its first version the engine is a single central point of trust, which is why it is not active. No bridge model is universally better.

STATE MACHINE

```
CREATED → WAITING_SOST → SOST_SEEN / CONFIRMING → SOST_FINAL (30 conf.)
→ AUTHORIZED (EIP-712, 30 min) → WAITING_ETHEREUM → MINTED → COMPLETED
exceptions: EXPIRED · PAUSED (burns continue) · RECOVERY_PENDING / REFUND_ELIGIBLE
            → REFUNDED (revoke on Ethereum first) · SECURITY_HOLD (reorg, inconsistency)
never REFUNDED and MINTED for the same deposit · each (txid, vout) consumed once
```

Custody target with V17 (#40,000) native multisig: a **3-of-5 Bridge Vault**, with spending keys separate from the Ethereum mint-authorization key. Status: **not audited**; contracts and engine are development prototypes. Full explanation, state names, comparison and worked examples: wSOST & Gold Vault.

SOST/EUR has **no approved methodology**: while it is absent the contract refuses every mint and the engine every request. The informational gold reference (0.8886707657 mg per SOST) is not a market price and is never used for fees.

TRUST — STATED PLAINLY

Ethereum cannot read SOST, so in the current prototype backing is asserted by the operator's authorizer keys: a **central point of trust**. A compromised authorizer key could issue unbacked wSOST up to the per-mint, daily and exposure caps until reconciliation (BALANCED / PENDING / CRITICAL) pauses minting — this is why the design is **not ready for production**. Native SOST multisig is not active (planned for V17), so today the Bridge Vault could only be a single cold key with offline releases. A refund requires a final on-chain revoke and the expiry of every issued authorization, so nobody can keep both. Stronger options under study: SOST header relay with Merkle inclusion proofs, succinct validity proofs, delay/veto windows for large mints, threshold signing and native multisig.

> **BLOCKING SECURITY REQUIREMENT**
>
> A stolen verifier key must **NOT** allow minting wSOST without previously locked native SOST. Before production there must be a demonstrable property MINTED_WSOST <= VERIFIED_CONFIRMED_NATIVE_SOST_LOCKED, and the system must **fail closed** (no mint) whenever it cannot be demonstrated. The current prototype does not meet this yet.

OPEN BLOCKERS (PUBLIC)

1. non-discretionary native reserve: no arbitrary withdrawal by its signers (P1-P5)
2. entry release authority: a stolen verifier key must not release unbacked wSOST
3. entry claim liveness: the reserve always claims before the user's deadline
4. secure native custody with V17 multisig (3-of-5 target)
5. testnet proof of concept (entry, exit, refunds, replay, reorg, concurrency, theft)
6. approved SOST/EUR valuation method
7. reliable XAUT price oracle
8. external security audit
9. legal / regulatory review

GOLD RESERVE V2 POWERS

withdraw(address)		does not exist · issues nothing · provides no liquid
pause	2-of-5	moves nothing
critical rescue	4-of-5 + delay	only to the predefined Quarantine Vault
final migration	4-of-5 guardians + YES ≥ ceil(2/3 × full eligible census) + timelock + exact successor vault named in the frozen proposal	once gover
guardian recovery	4-of-5 + timelock, cancellable by any 2; never moves assets	

SOST is not individually gold-backed: the reserve is restricted external ecosystem wealth with no individual right of redemption, parity, yield or guarantee.

GOVERNANCE CONSTITUTION

census	eligible identities at the snapshot (mining + NODE_BIND + heartbeats); one identity
vote	self-transfer from the identity with capsule SOSTGOV1|<proposal>|S<snapshot>|YES|N (28 of 80 bytes); last valid vote confirmed before the close counts
approval	YES ≥ ceil(2/3 × TOTAL census) — not of those who voted; ABSTAIN is never YES
example	census 12 → 8 YES needed · 8/2/2 approved · 7 YES + 5 no vote → not approved
finality	result final only after the close block is buried
phase	advisory first; binding only after objective maturity criteria (under research)

SOST votes do not execute Ethereum state: guardians do. Ignoring a valid vote would be publicly provable, not technically impossible — hence four layers: guardians, vote, timelock and destination-restricted contract. SOST and wSOST are never described as gold-backed, pegged, floored or redeemable.

HISTORICAL / SUPERSEDED DESIGN (kept for the record, never operational): the Gold → wSOST desk (selling existing treasury wSOST for XAUT at auction/TWAP prices), funding Gold Reserve V2 through that sale, 2-of-3 federated observer attestations with a relayer paying the mint gas, and the WSOST:0x… deposit capsule. It is not operational and is no longer the target architecture.

// POSITIONING

SOST vs other PoW chains

SOST is not positioned as a Bitcoin clone. It combines Bitcoin-like timing discipline, CPU-heavy original work, mathematical scarcity, and a constitutional reserve design. The four tables below place SOST against six well-known PoW chains so the reader can see exactly where the design overlaps and where it diverges.

Snapshot refreshed **2026-10-06 UTC** — a point-in-time reading, not updated on every page load. The SOST row is live from the explorer; the six comparison chains are current heights as of that date. Drift = current height - expected, where **expected = (now - chain genesis) ÷ canonical target spacing** (computed two-phase for Monero and Zcash, which changed target). Block heights and schedule drift change every minute; **see the live explorer at sost-explorer.html** for the

current SOST height. Sources for the comparison columns: Blockchain.info (BTC), Blockchair (LTC, DOGE, BCH, ZEC), moneroblocks.info (XMR).

1. Schedule — ahead, behind, or on time

CHAIN	TARGET / BLOCK	HEIGHT NOW	EXPECTED	DRIFT	STATUS
Bitcoin	600 s (10 min)	970,202	~933,800	+36,400	Ahead ~253 days
Litecoin	150 s (2.5 min)	3,190,417	~3,155,500	+34,900	Ahead ~61 days
Dogecoin	60 s (1 min)	6,546,187	~6,749,600	-203,500	Behind ~141 days
Monero	120 s (60 s before 2016)	3,778,247	~3,781,100	-2,900	Essentially on time
Bitcoin Cash	600 s	971,773	~933,800	+37,900	Ahead ~264 days
Zcash	75 s (150 s pre-Blossom)	3,508,520	~3,526,300	-17,800	Behind ~15 days
SOST	600 s	29,530	~29,511	+19	Essentially on time

Reading: chains with bulk retarget (BTC, LTC) accumulate large positive drift because their hashrate has grown exponentially faster than two-week recalibration can correct. BCH switched to ASERT in 2020 and the drift growth flattened. Dogecoin's drift is negative because its merge-mining economy with Litecoin and DigiShield retarget combine to produce blocks slower than its 1-minute target. Monero, Zcash and SOST track their schedule within a fraction of a day — the property delivered by per-block retarget.

CAPACITY & SCALABILITY

Transaction capacity & scalability

Why **~7 TPS** does **not** mean 7 transactions per block.

Why ~7 TPS ≠ 7 transactions per block

VERIFIED PARAMS

SOST is a Proof-of-Work, UTXO blockchain. Throughput comes from three verifiable consensus parameters: **block size ≈ 1 MB** (`MAX_BLOCK_BYTES_CONSENSUS = 1,000,000`), **target spacing ≈ 600 s** (`TARGET_SPACING = 600`), and a UTXO transaction whose serialized size varies with its inputs/outputs/data (a minimal 1-in / 1-out transfer ≈ **170 bytes**). A block may hold up to 65,536 tx by consensus; the default miner template caps selection at 4,096.

~7 TPS does not mean each block has 7 transactions. It means: thousands of transactions fit in one block ÷ ~600 seconds ≈ average sustained TPS.

THE FORMULA

```
TX_per_block  ≈  usable_block_bytes / average_TX_bytes
TPS           ≈  TX_per_block / target_block_spacing
```

Worked example — if an average simple transaction were ~250 bytes:

```
1,000,000 bytes / 250 bytes  ≈  4,000 transactions / block
4,000 / 600 seconds          ≈  6.7 TPS
```

Capacity by transaction size

APPROXIMATE

AVERAGE TX SIZE	APPROX TX / BLOCK	APPROX TPS
170 bytes	~5,800	**~9.7**
200 bytes	~5,000	~8.3
250 bytes	~4,000	~6.7
300 bytes	~3,300	~5.5
500 bytes	~2,000	~3.3

Approximations, not a guaranteed maximum — they ignore header/coinbase overhead; transactions with more inputs/outputs or payload are larger. SOST can include several thousand ordinary transactions per block.

The mempool & daily capacity

THROUGHPUT

A full block does not drop transactions — the surplus waits in the mempool for the next blocks (fee-rate ordered, policy-limited):

```
10,000 transactions waiting
        ↓
  BLOCK 1  ~4,000 TX   →  ~6,000 remain
        ↓
  BLOCK 2  ~4,000 TX   →  ~2,000 remain
        ↓
  BLOCK 3  remaining TX
```

DAILY CAPACITY (ILLUSTRATIVE)

SUSTAINED TPS	TRANSACTIONS / DAY
5 TPS	≈ 432,000
7 TPS	≈ 604,800
10 TPS	≈ 864,000

Theoretical/sustained illustrations — not current usage. One UTXO transaction can also pay many destinations at once (batching), so economic throughput can exceed one-payment-per-transaction.

Is 1 MB too small? What scaling would cost

TRADE-OFFS

For a young PoW network, ~1 MB is a conservative choice: low bandwidth, fast propagation, slow disk growth, cheap/accessible full nodes — all favouring decentralization.

> Unused block capacity has no value by itself. The question is how much throughput the network actually needs while keeping node operation accessible and independently verifiable.

WHAT A BIGGER BLOCK WOULD COST (≈250-BYTE TX)

BLOCK SIZE	APPROX TPS	CHAIN GROWTH (FULL BLOCKS)
1 MB	~6.7 TPS	~52–53 GB/yr
2 MB	~13.3 TPS	~105 GB/yr
4 MB	~26.7 TPS	~210 GB/yr
8 MB	~53 TPS	~420 GB/yr

Growth figures are maxima if blocks were continuously full (~52,560 blocks/yr). Bigger blocks → more bandwidth, storage and propagation time, higher stale/orphan risk, costlier nodes, centralization pressure.

AND A SHORTER BLOCK TIME?

BLOCK TIME	APPROX TPS (1 MB)	A 288-BLOCK WINDOW IS…
600 s (now)	~6.7 TPS	288 blk ≈ 2 days
300 s	~13.3 TPS	288 blk ≈ 1 day
120 s	~33 TPS	288 blk ≈ 9.6 h
60 s	~67 TPS	288 blk ≈ 4.8 h

Changing block time is not a simple performance switch. It would ripple through cASERT/difficulty, propagation, stale/orphan behaviour, forks/reorgs, SACS, DTD & Jackpot windows, emission timing, timestamps, wallets, miners, explorer math — and every rule expressed in block counts. It would require a hard fork plus a serious redesign and test.

Honest comparison — top 10 PoW + non-PoW contrast

SOURCED & LABELLED

TPS figures across blockchains are not directly comparable. Transaction complexity differs (UTXO vs account), smart-contract execution differs, vote/consensus transactions may be counted differently, block/slot architectures differ, and theoretical, benchmarked and observed TPS are different metrics. This is not a marketing leaderboard.

TOP PROOF-OF-WORK CHAINS (SOST'S CAMP)

NETWORK	CONSENSUS / MODEL	BLOCK / SLOT	APPROX TPS	HOW MEASURED	TRADE-OFF
SOST	PoW · ConvergenceX (CPU) · UTXO	~1 MB / ~600 s	**~5–10**	protocol-capacity estimate — **not** benchmarked	security + decentralizatio
Bitcoin	PoW · SHA-256d · UTXO	~1–4 MB wt / ~600 s	~7 (up to ~27 batched)	practical, weight-dependent	maximum security; ASIC
Litecoin	PoW · Scrypt · UTXO	~1 MB / ~150 s	~56 (capacity)	capacity (faster blocks)	faster settlement; ASI
Dogecoin	PoW · Scrypt (merged) · UTXO	~1 MB / ~60 s	~33 (capacity)	capacity	fast/cheap; merge-mined
Bitcoin Cash	PoW · SHA-256d · UTXO	~32 MB / ~600 s	~100–200+ (capacity)	capacity (big blocks)	throughput → heavier nodes
Monero	PoW · RandomX (CPU)	dynamic / ~120 s	~a few (flexible)	dynamic block + privacy	privacy + CPU mining
Ethereum Classic	PoW · Etchash · EVM	gas-limited / ~13 s	~15 (gas-limited)	gas-limited	EVM + PoW
Zcash	PoW · Equihash · shielded	~2 MB / ~75 s	~27 (capacity)	capacity (shielded heavier)	zk privacy
Dash	PoW · X11 · UTXO	~2 MB / ~150 s	~28–56 (InstantSend)	capacity / InstantSend	instant & private send
Kaspa	PoW · kHeavyHash · BlockDAG	~1 block/s	~hundreds–1,000s	high block rate (GHOSTDAG)	high PoW TPS → heavier bandwidth
Ravencoin	PoW · KAWPOW · UTXO	~1 min	~tens (capacity)	capacity	asset issuance

FOR CONTRAST — HIGH-THROUGHPUT NON-POW

NETWORK	CONSENSUS / MODEL	APPROX TPS	HOW MEASURED	TRADE-OFF
Ethereum (L1)	PoS · account/EVM	~15 (L1)	gas-limited; scales on L2	decentralized, L1-limited
TRON	DPoS (27 super reps)	~2,000 **(claimed)**	claimed; real throughput **questioned lower**	few producers → more centralized
Solana	PoH + PoS · parallel	~65,000 theo. / ~1–3k real	theoretical vs observed non-vote	very high TPS; high-spec validators

The high-TPS chains reach those numbers with a very different, more centralized architecture. **SOST deliberately sits in the Bitcoin camp** — trading raw throughput for security, decentralization and CPU-fair mining anyone can verify.

Where SOST stands & future research

FUTURE · NOT CONSENSUS

> The objective is not the highest TPS number — it is the highest **useful** throughput SOST can safely support without compromising the ability to run and independently verify the network.

Current priority order: **security → decentralization → node accessibility → reliability**, before raw TPS.

> **Future scalability is RESEARCH, not consensus.** No announced change, no activation height, no commitment to larger blocks or shorter block times. Possible research: transaction batching, more efficient serialization, UTXO consolidation, native-asset efficiency, and block-size / block-time lab experiments measuring sustained & peak TPS, propagation p50/p95/p99, CPU, RAM, disk, bandwidth, mempool, stale/orphan rate, sync and the effect on low-resource and distant nodes.

Verified from source: consensus_constants.h (1,000,000 B), params.h (600 s), transaction.cpp (input 133 B, output 30 B+), mempool.h (template 4,096). External figures are approximate and labelled by how measured; capacity ≠ current utilization, theoretical ≠ benchmarked. Sources: ethereum.org; TRON network-claimed ~2,000 TPS (real throughput questioned in public reporting); Solana docs/

explorers (~65,000 theoretical vs ~1,000-3,000 real non-vote); other PoW figures are approximate protocol capacities.

2. Emission and supply

CHAIN	BLOCK REWARD (NOW)	NEXT CHANGE	FINAL SUPPLY	NOTES
Bitcoin	3.125 BTC	2028 → 1.5625	21,000,000 BTC	Halvings every 210k blocks (~4 yr)
Litecoin	6.25 LTC	~2027 → 3.125	84,000,000 LTC	Halvings every 840k blocks (~4 yr)
Dogecoin	10,000 DOGE (fixed)	never	No cap	Perpetual inflation, ~5.26 B DOGE / yr
Monero	~0.6 XMR (tail)	fixed since 2022	~18.4 M + tail	Tail emission 0.6 XMR/block forever from June 2022
Bitcoin Cash	3.125 BCH	2028 → 1.5625	21,000,000 BCH	Inherited BTC's halving calendar at the 2017 fork
Zcash	1.5625 ZEC	2028 → 0.78125	21,000,000 ZEC	Halvings every 4 yr
Σ SOST	**7.85100863 SOST** (epoch 0)	block 131,553 (~2.5 yr)	**4,669,201 SOST**	Continuous decay $q = e^{-1/4} \approx 0.7788$ per epoch

Bitcoin and its derivatives use **discrete halvings**. Monero and SOST use **continuous decay**. Doge does not decay at all (perpetual inflation). Monero's tail emission means its supply has no asymptote — only a curve that flattens to a constant absolute issuance.

3. PoW algorithm and difficulty retarget

CHAIN	POW ALGORITHM	DOMINANT HARDWARE	RETARGET	NOTABLE DESIGN
Bitcoin	SHA-256d (double SHA-256)	ASIC	Bulk every 2,016 blocks (~14 d)	The original. Stateless flat hash.
Litecoin	Scrypt	ASIC	Bulk every 2,016 blocks (~3.5 d)	Scrypt holds a 128 KB working buffer; ASIC-friendly since 2014.
Dogecoin	Scrypt (merge-mineable with LTC)	ASIC via LTC	DigiShield (per-block) since Feb 2014	Security tied to Litecoin's merge-mined hashrate.
Monero	RandomX	CPU (deliberate)	LWMA (per-block, 720-block window)	CPU-oriented VM with JIT execution of randomised bytecode programs and a multi-MB-class dataset/cache plus per-execution scratchpads. ASIC-hostile by construction.
Bitcoin Cash	SHA-256d (same as BTC)	ASIC	ASERT (per-block) since Nov 2020	Per-block since the post-fork DAA bug of 2017.
Zcash	Equihash ⟨200, 9⟩	ASIC (since 2018)	DigiShield-like (per-block, EMA)	Originally GPU-friendly memory-hard; ASICs reached the

CHAIN	POW ALGORITHM	DOMINANT HARDWARE	RETARGET	NOTABLE DESIGN
				chain by mid-2018.
Σ SOST	ConvergenceX	CPU + 8 GB RAM	cASERT (per-block, avg288 + E/B/H equalizer profiles)	Native, original algorithm — not a fork. Solves a 32×32 SPD linear system across 100k sequential rounds of integer-only gradient descent and proves the solution sits in a stable basin (Proof of Irreversible Convergence). 256-op per-block program. 4 GB dataset + 4 GB scratchpad per mining thread.

4. Properties at a glance

PROPERTY	BTC	LTC	DOGE	XMR	BCH	ZEC	SO
Hard cap	✓ 21M	✓ 84M	✗ infinite	✗ tail	✓ 21M	✓ 21M	✓ 4.6
Per-block retarget	✗ bulk	✗ bulk	✓ DigiShield	✓ LWMA	✓ ASERT	✓ DigiShield	✓ cASE
CPU-oriented PoW	✗	✗	✗	✓ RandomX	✗	✗ (was GPU)	✓ Converge
Memory-hard footprint	✗		✗		✗	~ partial	✓ 8 GB t

PROPERTY	BTC	LTC	DOGE	XMR	BCH	ZEC	SO
		~ 128 KB		√ RandomX (multi-MB class)			
Native (not a fork)	√	✗	✗	~ CryptoNight lineage	✗ BTC fork	✗ Equihash	√ Converge origin
Constitutional reserve	✗	✗	✗	✗	✗	✗	√ 50% g linke
Default-private transactions	✗	✗	✗	√ ring + stealth	✗	~ shielded opt-in	✗ (E2 layer DEX/Ta
Hard cap derived from a math constant	✗	✗	n/a	n/a	✗	✗	√ Feigen δ

Where SOST stands out among the chains in this table

1. The only PoW algorithm in this table that is genuinely native and not derived. SHA-256d was Bitcoin's original; Scrypt comes from Tarsnap; Equihash from Wagner's generalised birthday. Monero's RandomX is the closest other example of an originally designed algorithm, but it carries CryptoNight ancestry. ConvergenceX shares no core structure with any of these.
2. The only hard cap in this table derived from a universal mathematical constant (Feigenbaum's δ). The other caps are round numbers chosen for marketing alignment.
3. The only chain in this table with a **consensus-level allocation toward gold-linked reserves and custody mechanisms** in its history: 25% Gold Vault plus 25% PoPC until block 24,999 (now frozen and funding the DTD Jackpot; 0% of new emission since V15). This is not a peg, not a redemption claim, not a stablecoin mechanic. The reserve ratio is observable on-chain, not contracted.
4. Among the chains in this table, SOST combines per-block retarget with the largest mining-RAM footprint (8 GB) of any production CPU-oriented design. Monero is the only direct competitor on the CPU-friendly axis but uses a smaller working set.

5. Tightest schedule alignment in the comparison: SOST is currently within ~13 blocks of its target, helped by cASERT's 288-block average and dynamic delta cap (0% within ±15 s, then 0.5% / 1% / 2% / 3% by deviation).
6. Hard cap lower than the major PoW comparables in this table. With 4.67M coins capped, SOST's long-run scarcity is steeper than BTC, LTC, BCH, or ZEC. Whether that matters in market terms is a separate question.

Where SOST does not stand out — honestly

- **Default privacy:** Monero wins on transaction privacy by default. SOST has end-to-end encryption in DEX/Talk but the base ledger is transparent like Bitcoin.
- **Track record / hashrate / market depth:** SOST is days–months old; BTC is 17 years old. Comparing security-economic depth today would be misleading.
- **Liquidity and exchange listings:** SOST is pre-market — no exchange listings yet; exchange outreach is in progress. The chains in this table all have active markets.
- **External security audit:** none yet. Independent reviewers welcome.
- **Post-quantum signature safety:** the current scheme uses ECDSA over secp256k1 (canonical LOW-S) for transaction signatures and is not PQ-resistant; BIP-340 Schnorr is used only for SbPoW block-identity binding, not for spending. PQ migration is on the roadmap as a future hard fork.

Comparison block last updated alongside the snapshot above. Tables refresh on each whitepaper revision.